

Q-1) Write different levels in the memory hierarchy [1 mark]

Q-2) What is the relation between cost and access time when we go down the memory hierarchy? [1 mark]

Q-3) Explain the principle of locality of reference in the context of cache memories. [2 marks]

Q-4) Explain write-through and write-back policy [2 marks]

Q-5) The address is divided as follows for a direct mapped cache.

tag	index	Byte-offset
8	14	2

A) Give the distribution if the cache is converted to a 4-way set associative cache [1 mark]

tag	index	Byte-offset

B) For the above 4-way set associative cache, consider the address FF7FC. Identify the tag. Write the answer in hex format. [1mark]

C) Give 2 addresses in binary that map to same set as FF7FC. [1mark]

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Q-6) Calculate the parity bit for single error detection for the following data values **[2 marks]**

Data	Parity bit
31 ₁₀	
0100 1100 1111 0111	
1111 1010 11	
54 ₁₀	

Q-7) What is used to locate the page table for a given process? _____ **[1mark]**

Q-8) What is the full form of TLB? _____ **[1mark]**

Q-9) Virtual address is divided into which two parts? _____ **[1mark]**

Q-10) What is the typical size of a page? _____ **[1mark]**

Q-11) Consider a 2-way set associative cache having 256 lines and block size of 32 bytes. The memory address size is 32 bits. Find the number of bits tag, index and block offset

tag	index	Byte-offset

Write the answer in the given answer sheet for the following questions (Q-12 to Q-14)

Q-12) Consider a cache system having miss rate as 0.3, miss penalty as 20 cycles and hit time is 2 cycles. Calculate the Average Memory Access Time (AMAT) in the presence of the cache? **[2 marks]**

Q-13) Draw the diagram of a 2-way set associative cache showing hit/miss circuitry and data out. **[3 marks]**

Q-14) We have to design DRAM as a single chip with each location storing 4-bits. Total chip capacity is 2MB. Draw a block diagram of DRAM showing the address register, the data grid, row and column addressing mechanisms. **[3 marks]**